

École Nationale Supérieure des Arts et Métiers

Mechanical engineering design project

Integrated Mechanical Design and Verification of a 4.6 kW Urban Wind Turbine Drivetrain

Aerodynamic sizing, speed-increasing gearbox, shafts, bearings, housing, assembly and safety systems



Wind-turbine demonstrator and test environment

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Design point:	13 m/s nominal wind speed
Rated machine:	4.6 kW, four poles
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Abstract

This report consolidates the preliminary sizing, detailed gearbox design and structural verification of a small urban wind-turbine drivetrain rated at 4.6 kW. The proposed architecture combines a straight-bladed rotor of projected area 9.9 m^2 , a belt stage with ratio 1.13, a two-stage spur-gear speed increaser with tooth-count ratio 13.984, and a four-pole aluminium-frame electrical machine. At the nominal wind speed of 13 m/s, the incident wind power is 14.14 kW; a prescribed power coefficient of 0.40 gives 5.66 kW at the rotor and implies an efficiency of 81.3% between the aerodynamic extraction and the 4.6 kW machine target. The recomputed gearbox output speed is 1522 rpm, only 0.9% below the original 159.8 rad/s target.

The gear geometry closes kinematically with normal modules of 1.75 mm and 1.25 mm, centre distances of 153.1 mm and 100.6 mm, and tangential mesh loads of approximately 3.37 kN and 1.38 kN. Correct conversion from angular speed to revolutions gives bearing design lives of 408, 1488 and 5713 million revolutions for the three shafts over 62 400 h. The corresponding basic dynamic load requirements are recalculated for all six bearing positions.

The structural audit is decisive. Reported Abaqus results give maximum von Mises stresses of 466.9, 116.5 and 16.1 MPa for the input, intermediate and output shafts. Against an allowable stress of 200 MPa, the input shaft fails with utilisation 2.33. Reported displacements of 1.20 and 1.49 mm also exceed the adopted $L/800$ criterion for the input and intermediate shafts, whereas the output shaft passes. Moreover, the analytical intermediate-shaft deflection of 3.5×10^{-5} mm differs from the finite-element value by more than four orders of magnitude. The present design is therefore a strong concept but not yet a released design: unit conventions, load application and boundary conditions must be reconciled, followed by mesh convergence and fatigue verification. This distinction between concept maturity and verified performance is the central engineering conclusion.

Keywords: small wind turbine; drivetrain; speed increaser; spur gears; shaft design; rolling bearings; finite-element analysis; interference fit; splash lubrication; design verification.

Nomenclature

Symbol	Definition	Unit
A	Rotor projected area	m^2
b	Gear face width	mm
C	Torque or bearing basic dynamic load rating, according to context	N m, kN
C_p	Rotor power coefficient	–
d, D	Inner and outer diameter	mm
E, G	Young and shear moduli	GPa
F_t, F_r	Tangential and radial gear forces	N
J	Polar or mass moment of inertia, according to context	$\text{m}^4, \text{kg m}^2$
K_u, K_v	First- and second-stage gear ratios	–
K_m	Overall gearbox ratio	–
L_{10}	Basic rating life at 90% reliability	rev
m_n	Normal gear module	mm
n, ω	Rotational speed	rpm, rad/s
P	Power or equivalent bearing load, according to context	W, kN
R_e	Material yield strength	MPa
V	Wind or pitch-line speed, according to context	m/s
Z	Number of gear teeth	–
η	Efficiency	–
ρ	Air density	kg/m^3
σ_{vM}	von Mises equivalent stress	MPa
θ'	Twist per unit length	rad/m

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1 Introduction

1.1 Engineering context

Small wind turbines intended for urban or peri-urban environments operate under constraints that are different from those governing utility-scale machines. Available space is limited, inflow is turbulent, acoustic emissions are visible to nearby users, and maintenance access is comparatively difficult. These constraints drive the design toward compact, quiet and serviceable hardware, but do not relax the basic requirements of structural integrity, overspeed protection and reliable power transmission. The design considered here follows the complete engineering sequence from system sizing to detailed gearbox design, structural verification and mechanical integration.

The mechanical chain is unusually instructive because it couples several disciplines. Rotor performance sets the low-speed torque. The belt and two gear meshes transform that torque into the speed required by a four-pole electrical machine. Gear geometry sets bearing forces; bearings and gear positions determine shaft bending; shaft deformation influences alignment and contact load distribution; the housing closes the load path; and lubrication and interference fits determine whether the assembly remains functional in service. Treating these tasks independently can yield individually plausible components that do not form a coherent machine. The present work therefore uses a system-level verification sequence.

1.2 Objective and research questions

The objective is to determine whether the proposed architecture constitutes a mechanically coherent 4.6 kW drivetrain and to identify the evidence still required before design release. Four questions structure the investigation:

1. Does the aerodynamic design point provide sufficient power and does the transmission reach the required machine speed?
2. Are the selected gear geometry, bearings and shaft diameters mutually consistent with the calculated loads and service life?
3. Do the analytical and finite-element structural results agree within an engineeringly credible range?
4. Are the housing, fits, lubrication and braking concepts compatible with manufacture, assembly and safe operation?

1.3 Evidence hierarchy

Dimensions, component selections, CAD geometry and simulation outputs are assembled here into one continuous calculation chain. Every quantity that can be reconstructed from the design inputs is independently recalculated. When an input is missing—for example a complete material certificate, gear quality grade, bearing catalogue rating, mesh convergence study or measured efficiency map—the analysis records the consequence and specifies the test or calculation needed to close the issue. This is consistent with the verification logic used in small-wind-turbine design standards [5] and with general mechanical design practice [1].

2 Design Basis and Integrated Method

2.1 Principal requirements

Table 1 gathers the consolidated design inputs. The gearbox fatigue and bearing calculations use a reference operating schedule of 10 h/d, six days per week, for 20 years, giving 62 400 h. A more severe availability target of 20 h/d is retained as a sensitivity case for components driven by calendar duty and number of starts.

Table 1 – Consolidated design basis.

Parameter	Engineering interpretation	Value
Nominal wind speed	Aerodynamic sizing point	13 m/s
Maximum wind speed	Braking and yaw load case	20 m/s
Electrical-machine target	Four-pole aluminium-frame LSES 112 MU	4.6 kW
Rotor projected area	Blade length multiplied by rotor diameter	9.9 m ²
Rotor power coefficient	Prescribed project value	0.40
Rotor speed at nominal wind	Low-speed-shaft design condition	10.1 rad/s
Belt ratio	Speed-increasing pulley stage	1.13
Gearbox ratio	Two-stage target / tooth-count closure	14 / 13.984
Service life	20 years, 10 h/day, 6 days/week	62 400 h
Shaft steel properties	Linear elastic analysis	$E = 210 \text{ GPa}$, $\nu = 0.30$
Fit steel properties	Interference calculation	$E = 200 \text{ GPa}$, $R_e = 600 \text{ MPa}$

2.2 Iterative design loop

The design uses an iterative sequence in which gear geometry, bearings, shafts and CAD packaging are repeatedly updated. Figure 1 expresses the central dependency structure: changing one gear width changes shaft span, housing width, bearing reactions and deflection. The verification loop is completed by explicit unit checks and acceptance ratios.

The verification sequence is therefore: (i) power and kinematic closure, (ii) mesh loads and gear geometry, (iii) bearing life, (iv) shaft torsion and bending, (v) finite-element correlation, and (vi) assembly, lubrication and safety review. Each step uses SI units internally; angular speed in radians per second is converted to revolutions per minute before bearing-life calculations.

3 Aerodynamic Design Point and Energy Conversion

3.1 Rotor geometry and nominal power

The preliminary design defines a rotor diameter $\Phi_r = 1.8 \text{ m}$, blade length $L_p = 5.5 \text{ m}$ and chord $c = 0.78 \text{ m}$. The resulting projected area is

$$A = L_p \Phi_r = 5.5 \times 1.8 = 9.9 \text{ m}^2. \quad (1)$$

Using $\rho = 1.30 \text{ kg/m}^3$ and the nominal wind speed $V_0 = 13 \text{ m/s}$, the incident wind power is

$$P_\infty = \frac{1}{2} \rho A V_0^3 = 14.14 \text{ kW}. \quad (2)$$

For the specified $C_p = 0.40$, the aerodynamic power becomes

$$P_r = C_p P_\infty = 5.66 \text{ kW}. \quad (3)$$

Reaching the 4.6 kW machine target therefore requires a combined downstream efficiency

$$\eta_{\text{req}} = \frac{4.6}{5.655} = 0.813. \quad (4)$$

This value is plausible as a design target but is not a measured efficiency. It must absorb aerodynamic derating not already represented by C_p , belt losses, two gear meshes, bearings, seals and the electrical-machine efficiency. The adopted rotor torque is 503 N m at 10.1 rad/s, corresponding to 5.08 kW. The 10.2% difference from Equation (3) indicates an implicit additional coefficient of about 0.90. The power flow should therefore be frozen in a single requirements table before procurement.

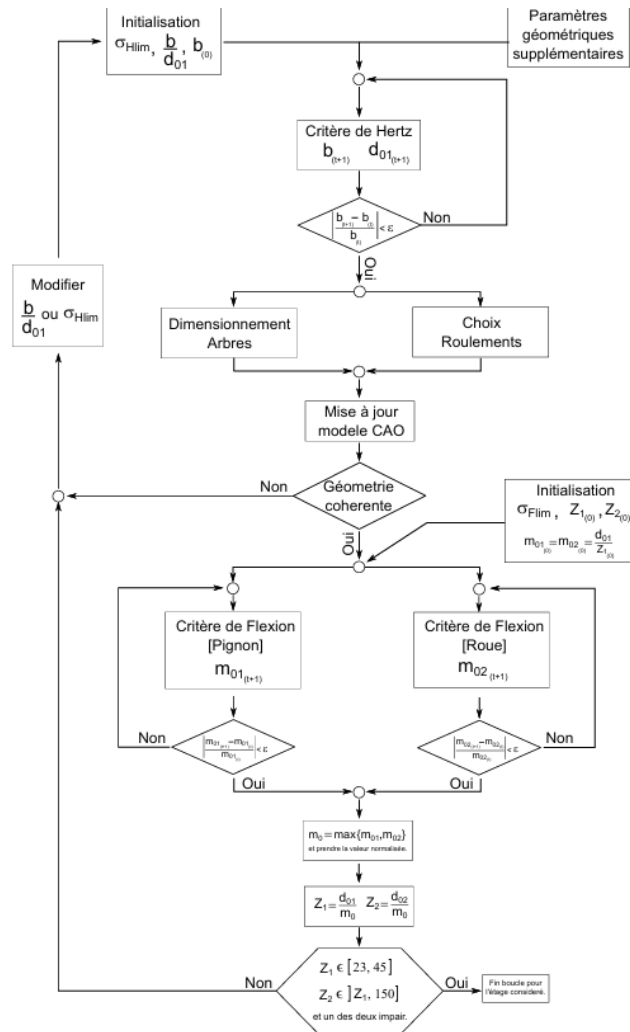


Figure 1 – Original iterative workflow for the speed-increasing gearbox.

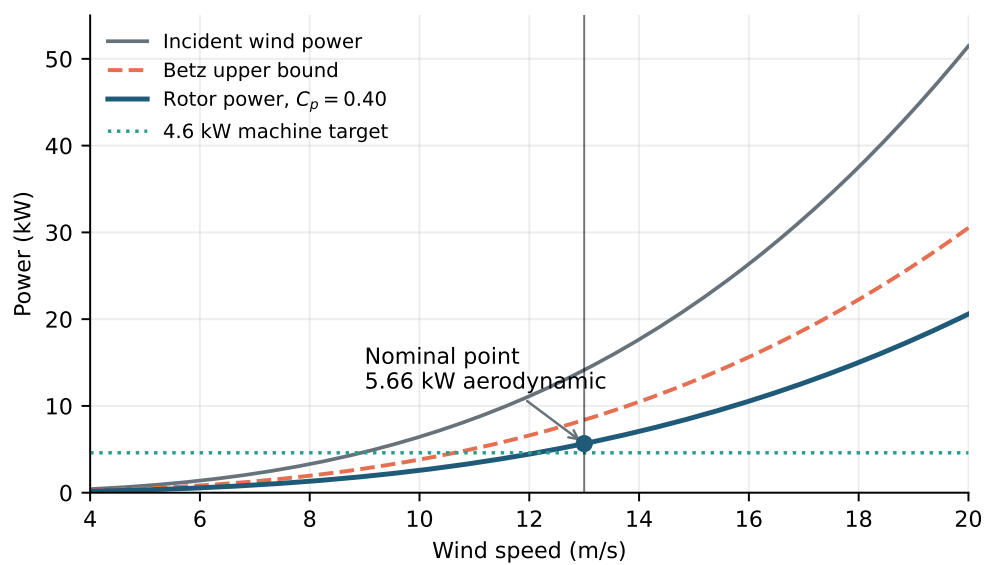


Figure 2 – Aerodynamic power envelope reconstructed from the design inputs. The generator line is a target, not a measured output curve.

3.2 Tip-speed ratio and operating plausibility

The rotor-speed relation $\omega_r = 2V\lambda/\Phi_r$ gives

$$\lambda = \frac{\omega_r \Phi_r}{2V_0} = \frac{10.1 \times 1.8}{2 \times 13} = 0.699. \quad (5)$$

A tip-speed ratio close to 0.7 can be consistent with a high-solidity, drag-dominated vertical-axis rotor, but the simultaneous use of $C_p = 0.40$ is optimistic and requires aerodynamic evidence. A wind-tunnel test or unsteady CFD study should provide the complete $C_p(\lambda)$ curve, starting torque and overspeed behaviour. Figure 3 shows the integrated demonstrator model used to establish packaging and support geometry.

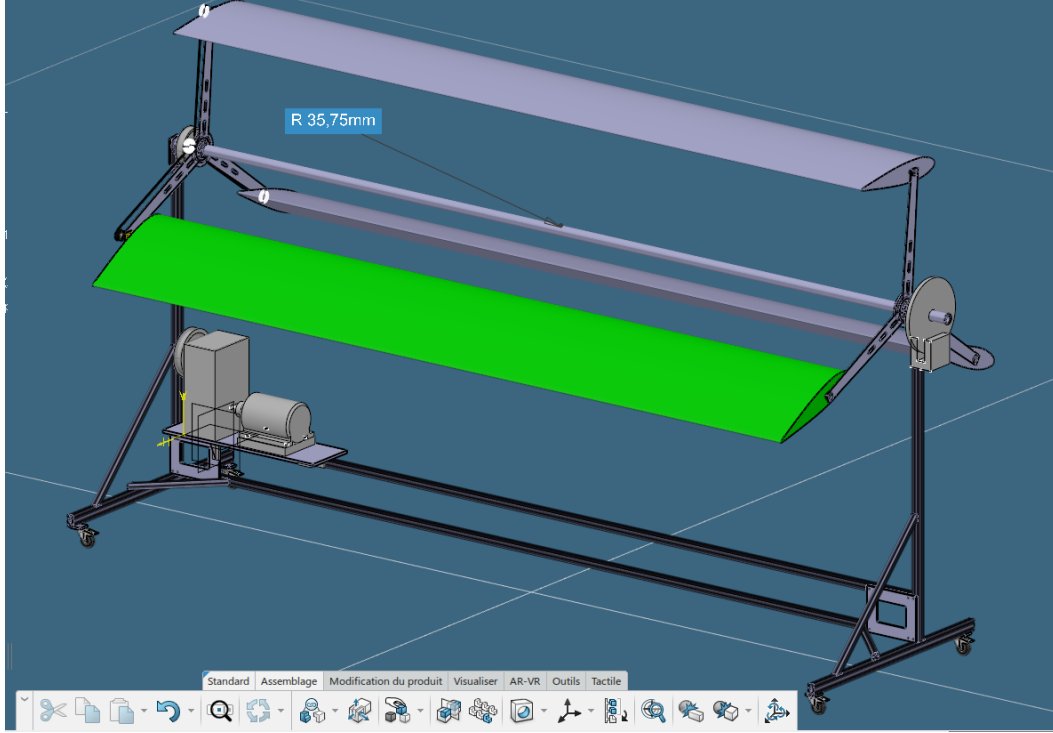


Figure 3 – Integrated turbine CAD model used for packaging of the rotor, drivetrain and support plate.

4 Drivetrain Architecture and Kinematic Closure

4.1 Selected architecture

The mechanical chain contains a belt stage followed by a two-stage spur-gear speed increaser. The belt raises rotor speed from 10.1 rad/s to approximately 11.4 rad/s. The gearbox then raises the speed to the four-pole machine. A two-stage layout was chosen to limit the ratio and size of each mesh while keeping the input and output on compatible sides of the installation.

The required total speed ratio follows directly from the rotor and generator operating points:

$$K_{\text{req}} = \frac{\omega_{\text{gen}}}{\omega_r} = \frac{161}{10.1} = 15.94. \quad (6)$$

With a belt ratio $K_f = 1.13$, the gearbox target is $K_m = K_{\text{req}}/K_f = 14.1$. Splitting this ratio over two meshes keeps each stage close to four and avoids the impractical pitch diameters, tooth-number constraints and overhung loads associated with a single-stage solution.

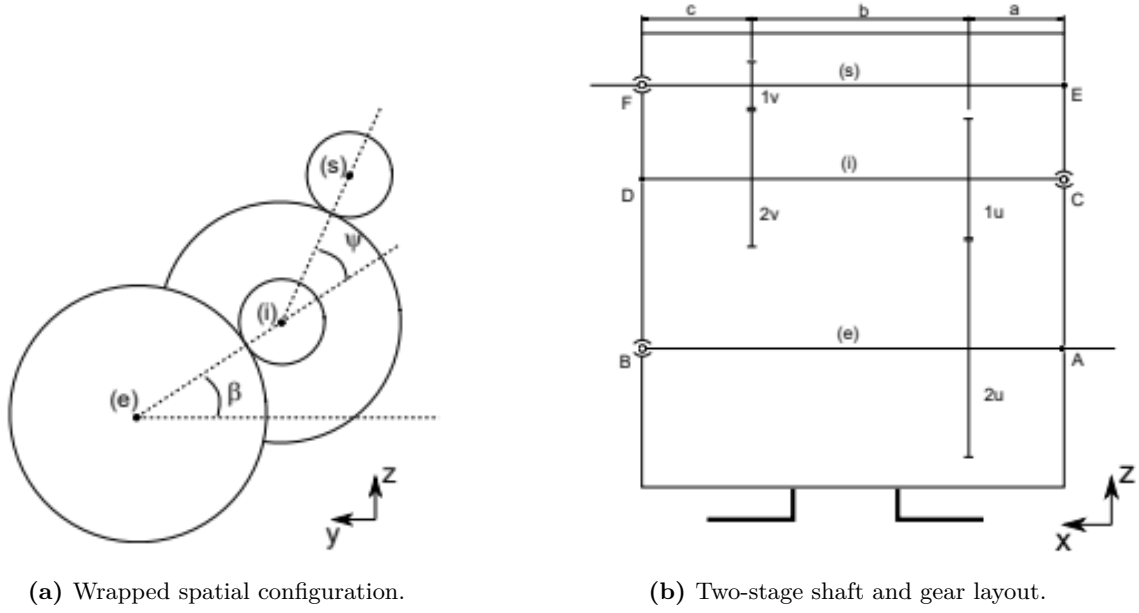


Figure 4 – Transmission topology used for the detailed calculations.

4.2 Tooth-count ratios

For a speed increaser, the larger wheel drives the smaller pinion at each stage. With the tooth counts in Table 2,

$$K_u = \frac{Z_{2u}}{Z_{1u}} = \frac{137}{38} = 3.6053, \quad (7)$$

$$K_v = \frac{Z_{2v}}{Z_{1v}} = \frac{128}{33} = 3.8788, \quad (8)$$

$$K_m = K_u K_v = 13.9841. \quad (9)$$

Including the belt stage gives $K_{\text{tot}} = 1.13 \times 13.984 = 15.802$, compared with the original target 15.94. The error is -0.87% , which is small relative to the aerodynamic variability but must be used consistently in speed, torque and reflected-inertia calculations.

Starting with $\omega_e = 11.4 \text{ rad/s}$ at the gearbox input, the tooth-count closure gives $\omega_i = 41.10 \text{ rad/s}$ and $\omega_s = 159.42 \text{ rad/s}$. The corresponding output speed is 1522 rpm. Relative to the 1500 rpm synchronous speed of a four-pole machine at 50 Hz, this is a generating slip of approximately -1.5% if the selected unit is operated as an induction generator. The exact torque-speed curve and permissible overspeed must be confirmed from the manufacturer.

4.3 Machine and auxiliary transmission components

The LSES 112 MU aluminium-frame machine was selected because the preliminary catalogue comparison indicated lower mass and sound level than equivalent cast-iron frames. The reported comparison gives 35–42 kg and approximately 49 dB for the aluminium alternatives, against heavier and noisier cast-iron options. These are catalogue values and should be checked at the actual generator operating point.

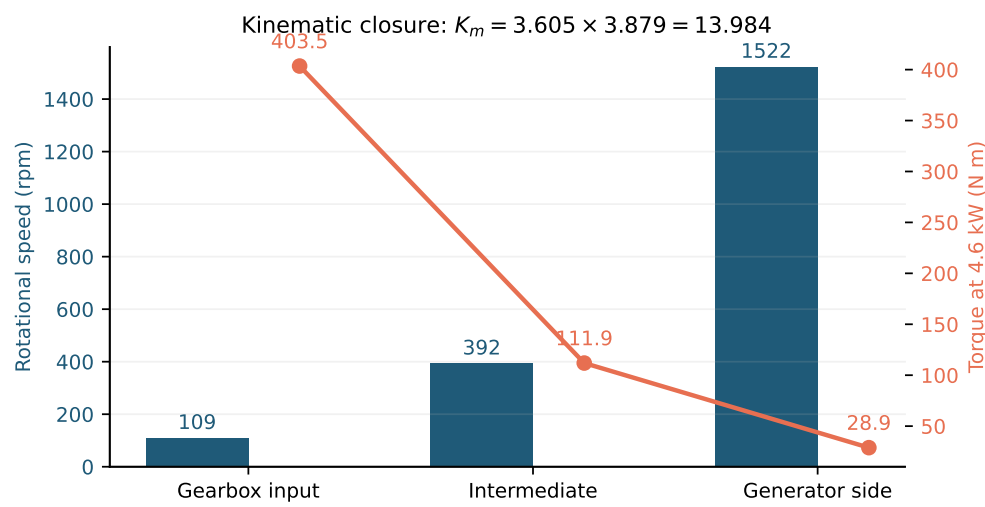


Figure 5 – Recomputed speed and torque along the two-stage gearbox at constant 4.6 kW transmitted power.

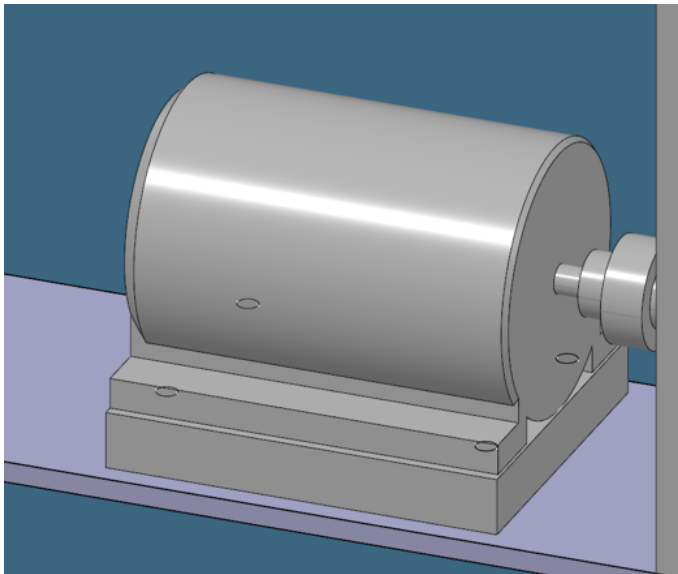


Figure 6 – CAD representation of the selected electrical machine and mounting feet.

4.4 Detailed belt-drive sizing

The continuous pulley solution uses a centre distance $e = 1.75$ m, preliminary ratio $K_f = 1.10$ and groove-geometry coefficient $g = 0.052$. The resulting driver and driven diameters are

$$d_{\text{driven}} = \frac{e(1-g)}{3(1-g+K_f)} = 270.0 \text{ mm}, \quad (10)$$

$$d_{\text{driver}} = \frac{eK_f}{3(1-g+K_f)} = 313.3 \text{ mm}. \quad (11)$$

The open-belt length and wrap angle on the smaller pulley follow from

$$L = 2e + \frac{\pi}{2}(D+d) + \frac{(D-d)^2}{4e} = 4.42 \text{ m}, \quad (12)$$

$$\theta = \pi - 2 \arcsin\left(\frac{D-d}{2e}\right) = 178.6^\circ. \quad (13)$$

At approximately 450 rpm, the belt speed is $v = \pi dn/60 \approx 6.36$ m/s.

Starting from a corrected transmitted power of 4.789 kW, a service factor $K_s = 1.50$ gives $P_s = 7.18$ kW. For a classical C-section belt with nominal single-belt rating $P_0 = 5.50$ kW, length factor $K_L = 1.10$ and wrap factor $K_\theta = 0.98$, the adjusted capacity is

$$P_a = P_0 K_L K_\theta = 5.93 \text{ kW}, \quad N_b = \frac{P_s}{P_a} = 1.21. \quad (14)$$

Two belts are therefore required. With friction coefficient $f = 0.50$, the Euler–Eytelwein relation adapted to the V-groove gives

$$\frac{T}{t} = \exp[3(1-g)f\theta] \approx 84, \quad (T-t)\frac{D}{2} = 503 \text{ N m}. \quad (15)$$

Solving these equations gives a tight-side tension $T \approx 3.25$ kN and slack-side tension $t \approx 38.6$ N, consistent with the rounded design values of 3.3 kN and 33.5 N. Catalogue diameters of 250 mm and 315 mm change the ratio and geometry; wrap, belt length, pretension and the radial shaft load must therefore be closed using the purchased components.

5 Gear Geometry and Load-Carrying Capacity

5.1 Final geometry

Table 2 consolidates the selected standard modules and tooth counts. The pitch diameter closure $d = m_n Z$ is exact for all four gears. Both pinions use $b/d = 1.25$, equivalently $d/b = 0.80$, matching the design parameter used in the iterative script.

Table 2 – Final spur-gear geometry.

Parameter	Pinion 1u	Wheel 2u	Pinion 1v	Wheel 2v
Normal module m_n (mm)	1.75	1.75	1.25	1.25
Tooth count Z	38	137	33	128
Pitch diameter d (mm)	66.50	239.75	41.25	160.00
Face width b (mm)	83.125	83.125	51.563	51.563
Centre distance (mm)	153.125		100.625	
Stage ratio	3.6053		3.8788	

The selected modules exceed the unrounded bending results of approximately 1.59/1.57 mm for stage u and 1.05/1.04 mm for stage v . Standardisation therefore adds a geometric margin and improves tooth robustness. The corresponding trade-offs are higher mass, greater sliding losses and increased noise. Those effects are acceptable only if tooth quality, backlash and housing stiffness are controlled.

5.2 Mesh forces

At the large driven wheel of stage u , $T = 403.51 \text{ N m}$ and $d = 239.75 \text{ mm}$. At the large wheel of stage v , $T = 110.55 \text{ N m}$ and $d = 160 \text{ mm}$. The tangential forces are

$$F_t = \frac{2T}{d}, \quad F_{t,u} = 3.37 \text{ kN}, \quad F_{t,v} = 1.38 \text{ kN}. \quad (16)$$

For a 20° pressure angle, the separating forces are $F_r = F_t \tan \alpha$, giving approximately 1.23 kN and 0.50 kN . These are the fundamental external loads for the shaft and bearing free-body diagrams.

5.3 Contact and bending criteria

The load-capacity algorithm follows the structure of ISO 6336 [8–10]. Surface durability is assessed by

$$\sigma_H = Z_H Z_E Z_\epsilon Z_\beta \sqrt{\frac{2TK_A K_V K_{H\alpha} K_{H\beta} u + 1}{bd_1^2} \frac{u + 1}{u}} \leq \frac{\sigma_{H,\text{lim}} Z_{NT}}{S_H}, \quad (17)$$

with load-distribution, application, dynamic, elasticity, contact-ratio and life factors. Tooth-root strength is assessed by

$$\sigma_F = \frac{F_t}{bm_n} Y_{FS} Y_\beta K_{F\beta} K_{F\alpha} K_A K_V \leq \sigma_{F,\text{adm}}. \quad (18)$$

The project used $K_A = 1.25$, $K_V = 1$, $K_{F\alpha} = 1.05$, $Y_\beta = 1$, $Y_{ST} = 2$ and $S_F = 2$, and obtained a face-load factor near $K_{H\beta} = 1.36$. Over $62\,400 \text{ h}$, the input, intermediate and output shafts accumulate 4.08×10^8 , 1.49×10^9 and 5.71×10^9 revolutions. With $Y_{NT} = (2 \times 10^6/N)^{0.02}$, the corresponding bending-life factors are 0.90 , 0.88 and 0.85 ; the adopted contact-life factor is approximately $Z_{NT} = 0.85$. The preliminary relation $\sigma_{F,\text{lim}} = \sigma_{H,\text{lim}}/4$ gives 218.8 MPa from a 875 MPa contact reference, or 206.3 MPa from 825 MPa . This ambiguity materially changes the margin, so the material grade and heat treatment must replace provisional stress values before release. A formal ISO 6336 calculation sheet should also record gear quality, reliability, rim thickness, surface finish and load spectrum.

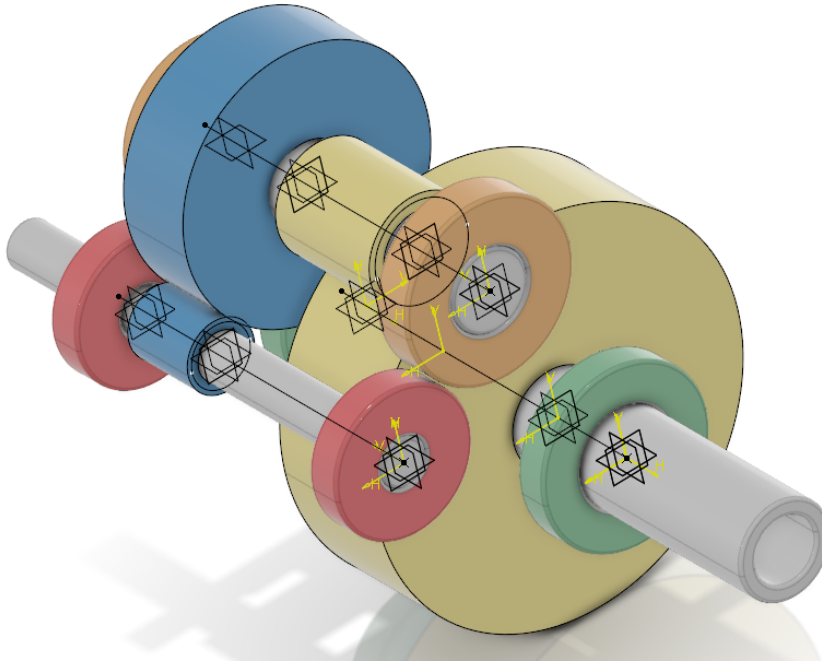


Figure 7 – Parametric CAD model of the final two-stage gearbox geometry.

5.4 Packaging result

The shaft-axis arrangement was first minimised analytically and then corrected in CAD to eliminate interference between root and tip circles. The selected angular placement was approximately 50° and 232° for the two mesh directions. CAD iteration reduced axial length from 290 mm to approximately 250 mm, a reduction of 13.8%. The reported gearbox envelope is 0.025 m^3 . This is a meaningful compactness improvement, but the final envelope should include seals, fastener access, oil ports and assembly clearances rather than gears and bearings alone.

6 Bearing Selection and Life Audit

6.1 Reaction-force model

The original free-body model resolves tangential and radial mesh forces in two orthogonal planes and adds belt tension at the gearbox input. Six bearing reactions, denoted A–F, support the three shafts. Figure 8 shows the geometry used to derive those reactions. In the first bearing pass, axial load was neglected and deep-groove ball bearings were selected. Later integration introduced self-aligning or double-row concepts at selected locations to tolerate alignment errors.

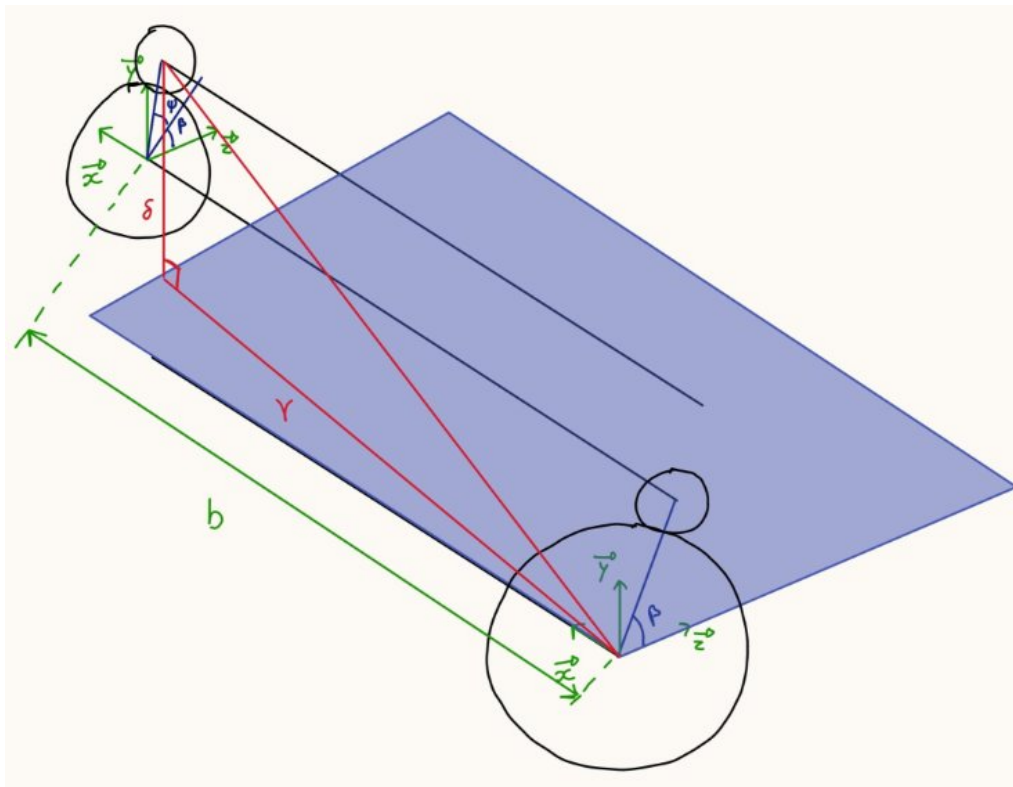


Figure 8 – Input-shaft free-body diagram and bearing-load convention.

For the input mesh, the tangential and radial tooth forces are $T_u = 2C_e/d_{2u}$ and $R_u = T_u \tan \alpha_0$. The small belt-line inclination is $\delta = (\pi - \theta)/2 \approx 0.71^\circ$. Static equilibrium in the two transverse

planes gives

$$Y_A = \frac{(b+c)(R_u \cos \beta - T_u \sin \beta) - (a+b+c+m)(T-t) \sin \delta}{a+b+c}, \quad (19)$$

$$Z_A = \frac{(b+c)(R_u \sin \beta + T_u \cos \beta) - (a+b+c+m)(T+t) \cos \delta}{a+b+c}, \quad (20)$$

$$Y_B = \frac{a(R_u \cos \beta - T_u \sin \beta) + m(T-t) \sin \delta}{a+b+c}, \quad (21)$$

$$Z_B = \frac{a(R_u \sin \beta + T_u \cos \beta) + m(T+t) \cos \delta}{a+b+c}. \quad (22)$$

Here a , b and c locate the bearings and gear plane, m is the pulley overhang, and β is the mesh orientation in the housing. The same force and moment balances are applied to the intermediate and output shafts, with equal and opposite mesh forces at each gear pair. This calculation yields radial equivalent loads of 4.0, 2.6, 2.9, 1.5, 0.5 and 1.0 kN at positions A–F. For purely radial loading,

$$P_i = \sqrt{Y_i^2 + Z_i^2}. \quad (23)$$

For ball bearings and 90% reliability, ISO 281 gives the basic relation [4, 6]

$$L_{10} = \left(\frac{C}{P}\right)^3, \quad C = PL_{10}^{1/3}, \quad (24)$$

where L_{10} is expressed in millions of revolutions.

6.2 Unit-corrected life

Bearing life must be counted in revolutions, not radians. The conversion from angular speed and operating hours is

$$L_{10}[10^6 \text{ rev}] = \frac{60n[\text{rpm}]L_h[\text{h}]}{10^6} = \frac{60^2 \omega L_h}{2\pi \times 10^6}. \quad (25)$$

For 62 400 h, the three shaft lives are 407.6, 1487.7 and 5713.3 million revolutions. Table 3 gives the corresponding required basic dynamic ratings.

Table 3 – Bearing-life recalculation from the reported loads.

Bearing	A	B	C	D	E	F
P (kN)	4.0	2.6	2.9	1.5	0.5	1.0
L_{10} (million rev)	407.6	407.6	1487.7	1487.7	5713.3	5713.3
C_{req} (kN)	29.7	19.3	33.1	17.1	8.9	17.9
Initial calculated C (kN)	55.0	35.4	61.1	32.0	32.0	16.3
Initial reference	6212	6212	6310	6310	6406	6406

The recalculation does not by itself approve a bearing. Manufacturer values of C , C_0 , limiting speed, internal clearance, seal drag and lubrication must be verified for the exact suffix. The preliminary references are generally conservative relative to the corrected requirements, but position F deserves an explicit catalogue check because its initial C value is below the recalculated requirement. The support-arm calculation gives an equivalent load of 1.294 kN and selects a 4307 ATN9 bearing with a required C near 42.7 kN; this selection should be propagated into the final bill of materials.

6.3 Low-speed rotor bearings

The rotor-support free body gives $Y_A = 1.246$ kN, $Z_A = 86$ N, $Y_B = -2.087$ kN and $Z_B = -86$ N. Hence

$$P_A = \sqrt{Y_A^2 + Z_A^2} = 1.249 \text{ kN}, \quad P_B = \sqrt{Y_B^2 + Z_B^2} = 2.089 \text{ kN}. \quad (26)$$

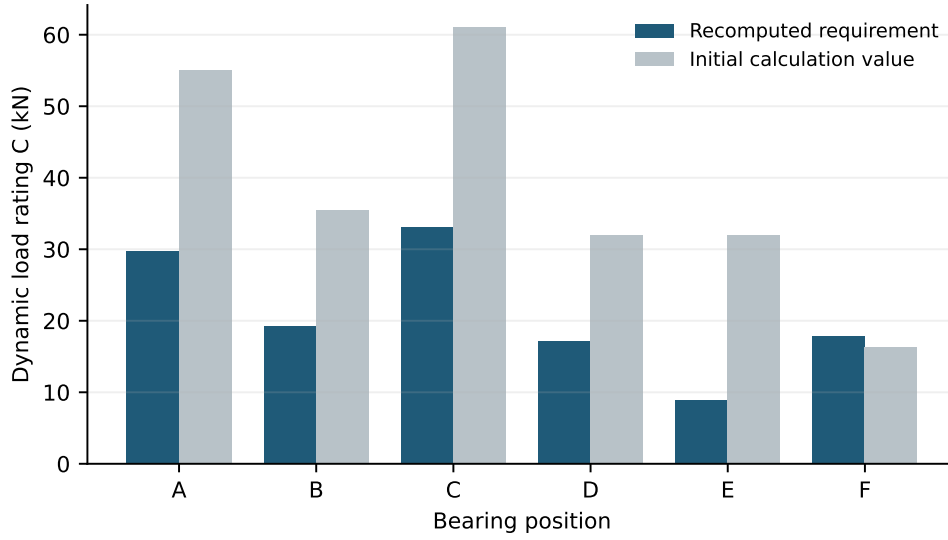


Figure 9 – Comparison of unit-corrected dynamic load requirements with the initial calculation values. Catalogue ratings must be checked separately.

At the rotor speed and specified life, the required dynamic ratings are approximately $C_A = 9.4$ kN and $C_B = 15.7$ kN. Candidate references 61815-2RS1 and 61915 were compared, and a common 61915 envelope was retained to simplify the support. The final reference still requires confirmation of its exact dynamic rating, seal arrangement, clearance after fit and corrosion protection.

7 Shaft Sizing and Analytical Structural Model

7.1 Torsional stiffness

The design rule limits twist per unit length to $\theta' \leq 0.5^\circ/\text{m}$. For a circular shaft,

$$\theta' = \frac{T}{GJ}, \quad J_{\text{solid}} = \frac{\pi D^4}{32}, \quad J_{\text{hollow}} = \frac{\pi(D^4 - d^4)}{32}. \quad (27)$$

Solving the twist criterion for diameter gives

$$D_{\text{min,solid}} = \left(\frac{32T}{\pi G \theta'} \right)^{1/4}, \quad D_{\text{min,hollow}} = \left[\frac{32T}{\pi G \theta' (1 - k^4)} \right]^{1/4}, \quad k = \frac{d}{D}. \quad (28)$$

For $G = 80$ GPa and $k = 0.60$, the nominal-torque minima are 51.0 mm, 35.6 mm and 25.5 mm for the input, intermediate and output shafts. Applying the adopted conservative load allowance raises the working minima to approximately 58.6, 41.0 and 29.3 mm; the selected 60 mm/40 mm hollow input shaft, 50 mm solid intermediate shaft and 30 mm solid output shaft therefore close the preliminary stiffness calculation. The direct checks are listed in Table 4. These calculations address pure torsion only; shoulder stress concentrations, keyways or fit pressures require separate fatigue checks.

Table 4 – Independent torsional-stiffness check of selected shaft diameters.

Shaft	Torque (N m)	Geometry (mm)	Twist (deg/m)	τ_{max} (MPa)
Input	403.51	$D = 60, d = 40$	0.283	11.86
Intermediate	110.55	$D = 50$ solid	0.129	4.50
Output	28.82	$D = 30$ solid	0.260	5.44

The upstream rotor shaft is treated separately because it carries the 503 N m aerodynamic design torque. For the selected tube, $D = 71.5$ mm and $d = 44.9$ mm,

$$J = \frac{\pi}{32}(D^4 - d^4) = 2.167 \times 10^{-6} \text{ m}^4, \quad \theta' = \frac{T}{GJ} = 0.166^\circ/\text{m}, \quad \tau_{\max} = \frac{T(D/2)}{J} = 8.30 \text{ MPa}. \quad (29)$$

The rotor shaft therefore passes the torsional-stiffness requirement with substantial elastic margin, before accounting for transverse blade and belt loads.

7.2 Beam model for bending

The analytical model treats each shaft as an Euler–Bernoulli beam between bearing centres. Gear forces and gear weights are concentrated at their application planes; shaft self-weight is distributed. Bearing moments are neglected. With $v(x)$ and $w(x)$ the deflections in the two transverse planes,

$$EI \frac{d^2 v}{dx^2} = -M_z(x), \quad EI \frac{d^2 w}{dx^2} = -M_y(x). \quad (30)$$

Piecewise integration over the intervals between bearings and gears provides slope and deflection. Continuity of displacement and slope is imposed at each load plane, and displacement is set to zero at the bearing idealisations. The reported intermediate-shaft maximum is only 3.5×10^{-5} mm, far below the $L/800 \approx 0.3$ mm criterion.

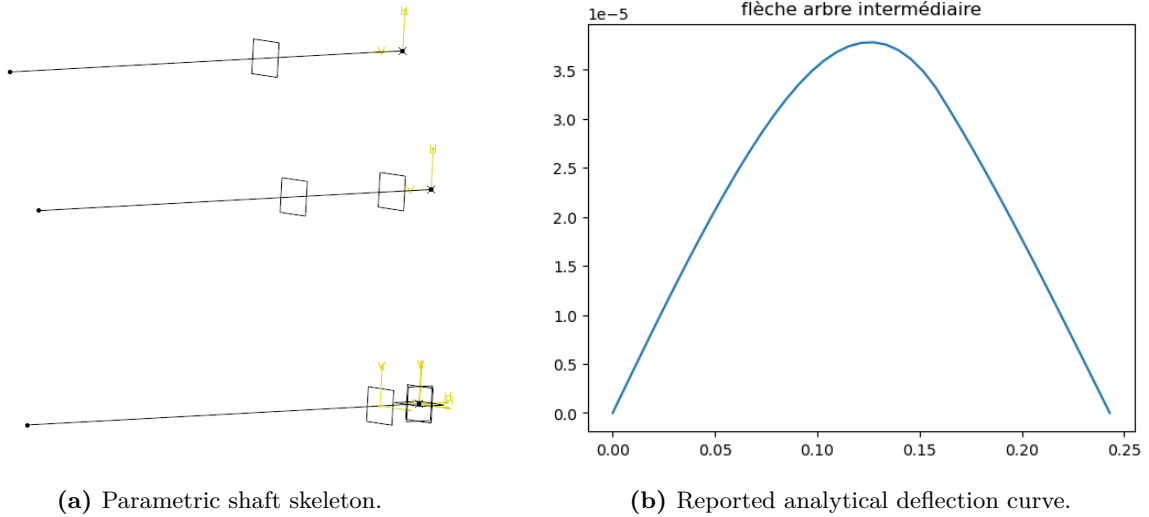


Figure 10 – Geometry and analytical bending model used before finite-element verification.

That result is not credible when compared with the Abaqus result of 1.49 mm. A factor greater than 4×10^4 cannot be attributed to normal differences between beam and finite-element models. Likely causes include inconsistent metre/millimetre units, incorrect area moment of inertia, omitted load components, or a boundary-condition mismatch. The analytical equations should be rebuilt in one consistent SI unit system and checked against a hand solution for a simply supported beam with one transverse point load.

8 Finite-Element Verification and Correlation

8.1 Model description

The three shafts were analysed in Abaqus as beam-like members with linear-elastic isotropic steel, $E = 210$ GPa and $\nu = 0.30$ [2]. Loads were applied at gear and pulley locations and supports represented the bearings. The input shaft was modelled as a hollow pipe using the minimum

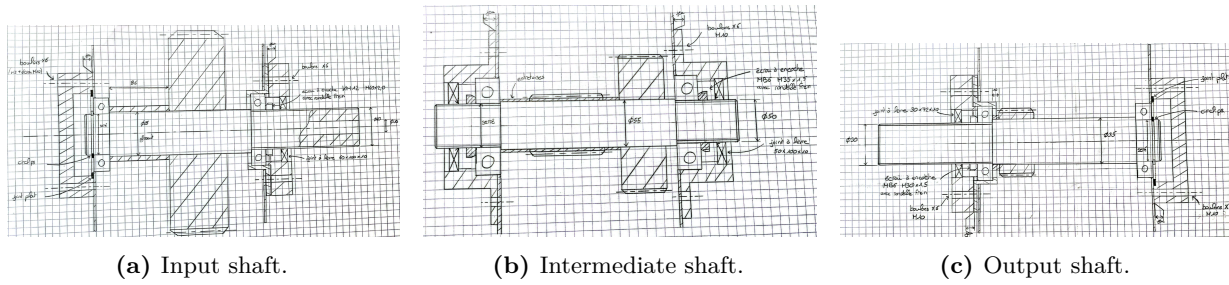


Figure 11 – Assembly drawings defining the stepped shafts, bearing seats and axial location features.

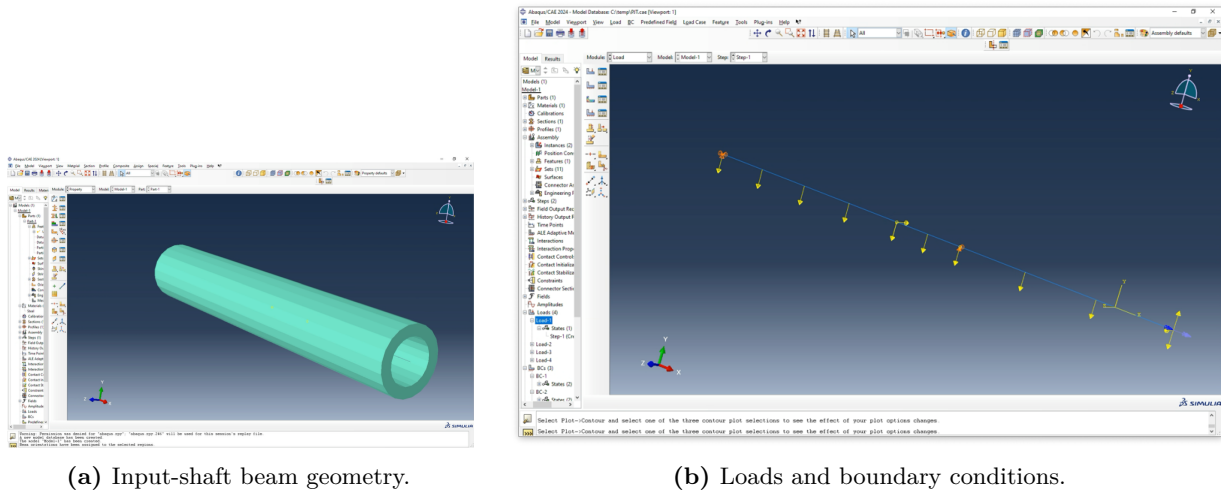


Figure 12 – Finite-element idealisation of the input shaft.

section, intentionally neglecting local shoulders in the first pass. Figure 12 illustrates the geometry and load application.

No mesh-density study, reaction-force balance or strain-energy comparison is documented. Consequently the results are treated as preliminary numerical evidence. A release-quality model should demonstrate equilibrium, confirm beam orientation and section assignment, report element type and size, and show convergence of peak stress away from idealised point loads.

8.2 Reported fields

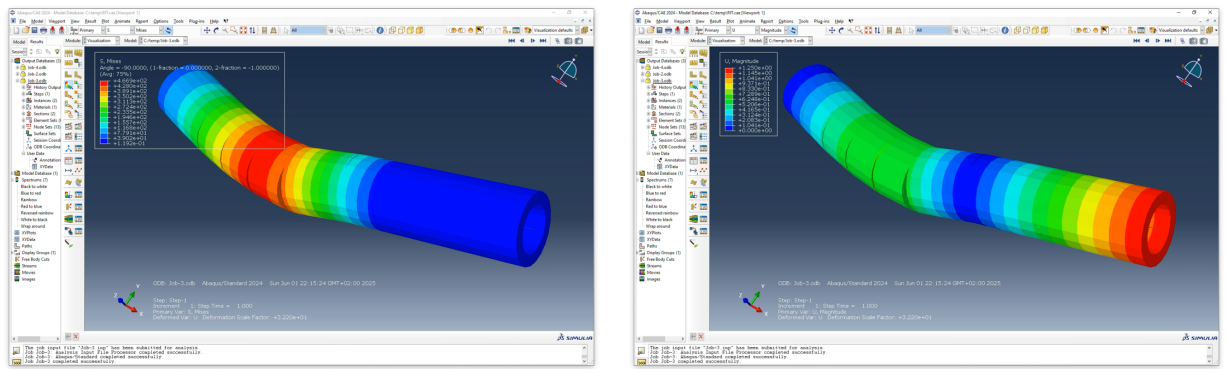
The input-shaft results are shown in Figure 13; the reported maximum von Mises stress is 466.9 MPa and the displacement is 1.20 mm. These values are significant because both exceed the project acceptance limits when a safety factor of three and $L/800$ deflection criterion are applied.

The intermediate and output fields are presented in Figure 14. Reported stresses are 116.5 MPa and 16.1 MPa; reported displacements are 1.49 mm and 0.0793 mm, respectively.

8.3 Acceptance audit

The interference-fit study uses $R_e = 600$ MPa and a safety factor $s = 3$, so the consistent static allowable is 200 MPa. For the input model length of 400 mm, $L/800 = 0.50$ mm. For a representative 245 mm span on the other two shafts, the limit is 0.306 mm. Table 5 and Figure 15 show the resulting utilisation.

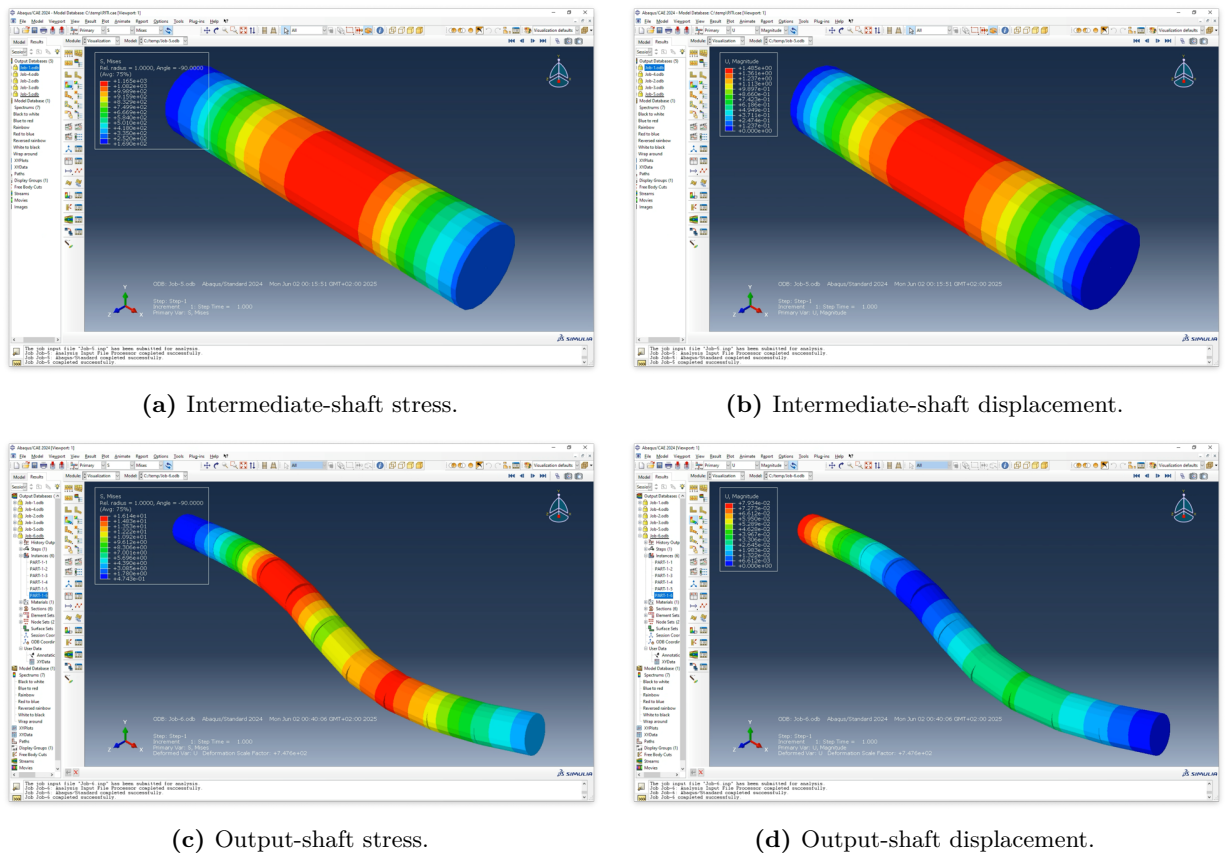
The correct conclusion is not that all shafts are verified. The input shaft requires redesign or correction of the numerical model; the intermediate shaft requires reconciliation of deflection units and boundary conditions; only the output shaft passes both checks on the available evidence. A practical redesign sequence is to verify loads and units first, then increase the input-shaft section or reduce span, add realistic bearing stiffness, and repeat the analysis with fillets and shoulders.



(a) von Mises stress field.

(b) Displacement field.

Figure 13 – Reported Abaqus results for the input shaft.



(a) Intermediate-shaft stress.

(b) Intermediate-shaft displacement.

(c) Output-shaft stress.

(d) Output-shaft displacement.

Figure 14 – Reported Abaqus fields for the intermediate and output shafts.

Table 5 – Structural acceptance audit of the reported finite-element results.

Shaft	σ_{vM} (MPa)	Stress util. /200	u_{max} (mm)	Defl. util. /(L/800)	Stress status	Deflection status
Input	466.9	2.33	1.20	2.40	Fail	Fail
Intermediate	116.5	0.58	1.49	4.87	Pass	Fail
Output	16.1	0.08	0.0793	0.26	Pass	Pass

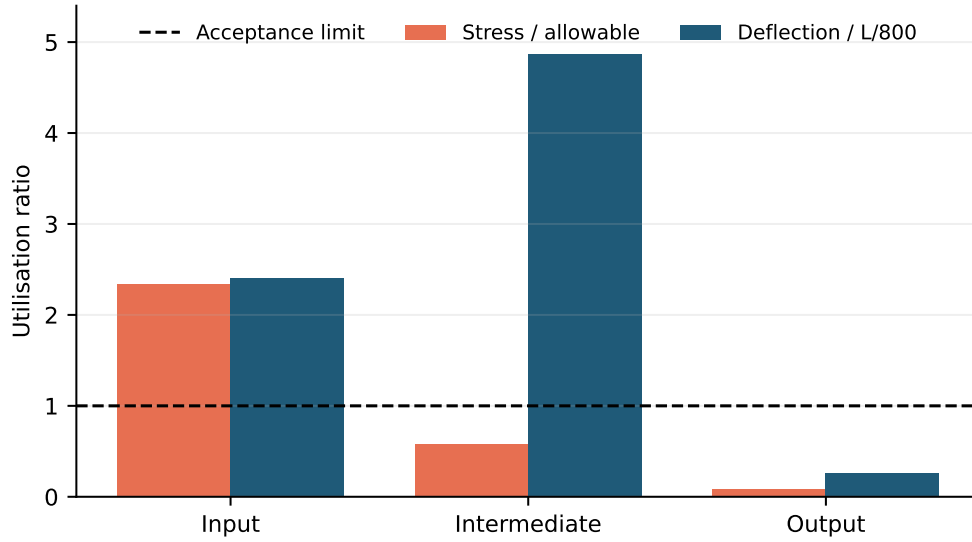


Figure 15 – Utilisation ratios derived from the reported finite-element results. Ratios above one do not meet the adopted criterion.

Gear alignment should be assessed from relative displacement at the mesh rather than only the global maximum displacement.

9 Housing, Support Arm and Assembly Technology

9.1 Housing and torque reaction

The housing consists of two sheet-metal half-shells joined by 40 fasteners and sealed by a flat gasket. Two service ports are required for filling and draining. The reported CAD material volume is 1510 cm³. Figure 16 shows three assembly states.

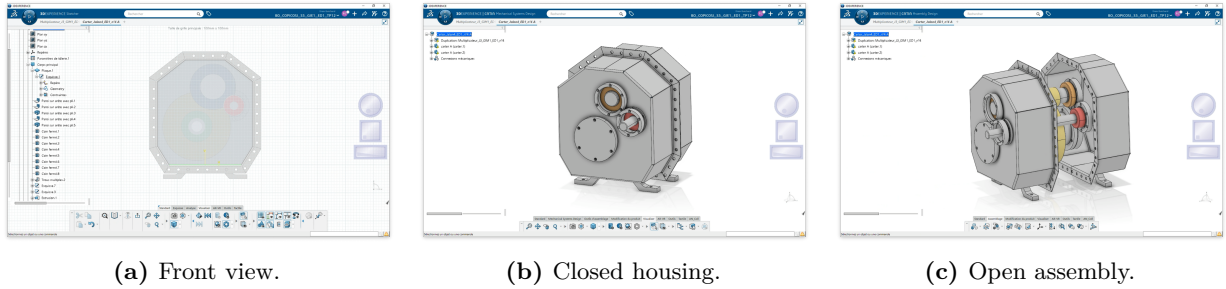


Figure 16 – Gearbox housing concept and integrated drivetrain.

The support arm reacts the external gearbox torque. The static model uses $C_e = 403.51 \text{ N m}$, $K_m = 14$, arm offsets $v_b = 75.4 \text{ mm}$, $h_b = 203 \text{ mm}$ and $\theta = 30^\circ$. The casing torque and arm force are

$$C_c = \frac{K_m + 1}{K_m} C_e = 432.3 \text{ N m}, \quad F_b = \frac{C_c}{v_b \sin \theta + h_b \cos \theta} = 2.03 \text{ kN}. \quad (31)$$

With a gearbox mass of 45.15 kg, the reported support reactions are $Z_G = -361 \text{ N}$, $Y_G = 1012 \text{ N}$, $M_{Y_G} = -1237 \text{ N m}$ and $M_{Z_G} = 304 \text{ N m}$. These values should be applied to the housing and support structure in a separate shell or solid finite-element model. A sheet thickness of 3 mm makes local bearing-seat stiffness, fastener preload and gasket compression especially important.

9.2 Bearing location and sealing

The input and output shafts use a locating/non-locating bearing philosophy to accommodate thermal growth, with a double-cap arrangement at the locating side, slotted nuts and locking washers for axial retention, and lip seals at rotating exits. The intermediate shaft uses a self-aligning concept to tolerate modest bending and assembly error. Inner rings are intended to be interference fitted because the load direction rotates relative to the shaft ring. This is consistent with rolling-bearing practice [4, 11], provided internal clearance after fitting is checked.

The drawings show a credible assembly sequence, but the housing thickness prevents reliable tapped M10 threads; through-bolts and nuts were therefore proposed. Before manufacturing, the design needs explicit fits for every bearing seat, a datum scheme for coaxial bearing bores, surface-finish specifications for seal tracks, axial end-float targets, and access for tightening slotted nuts. The exaggerated CAD clearances should be replaced by a tolerance stack.

10 Interference-Fit Design

10.1 Pressure and minimum interference

Gear-to-shaft joints are designed as steel-on-steel interference fits. The minimum interface pressure required to transmit torque is

$$p_{\min} = \frac{2Tk}{\pi\mu Ld^2}, \quad (32)$$

where $\mu = 0.10$ is the friction coefficient, $k = 2$ is the torque safety factor, L is engagement length and d is nominal diameter. For identical elastic materials, the radial-compliance relation is evaluated using shaft inner diameter d_i and hub outer diameter d_e , following the classical thick-cylinder treatment [1, 3]. Tolerance-zone extremes are then added to obtain maximum interference and pressure.

For wheel 2u, $T = 403.51 \text{ N m}$, $L = 83.125 \text{ mm}$ and $d = 60 \text{ mm}$. Substitution in Equation (32) gives

$$p_{\min} = \frac{2(403.51)(2)}{\pi(0.10)(0.083125)(0.060)^2} = 17.17 \text{ MPa}. \quad (33)$$

The elastic shaft–hub compliance converts this pressure into a minimum diametral interference of $114.7 \mu\text{m}$. Adding the H6/r6 tolerance extremes gives $152.7 \mu\text{m}$ maximum interference and 22.9 MPa maximum interface pressure. The same sequence is applied to all four joints.

Table 6 – Interference-fit results reported for the four gears.

Joint	T (N m)	d (mm)	L (mm)	$p_{\min}$ (MPa)	$\Delta_{\min}$ (μm)	$\Delta_{\max}$ (μm)	$p_{\max}$ (MPa)	$\sigma_{h,\max}$ (MPa)	ΔT ($^{\circ}\text{C}$)	Fit
Pinion 1u	110.55	50	83.125	6.8	55.2	87.2	10.7	44.8	404	H6/p6
Wheel 2u	403.51	60	83.125	17.2	114.7	152.7	22.9	42.3	427	H6/r6
Pinion 1v	28.82	30	51.56	7.9	31.9	57.9	14.3	54.9	591	H6/p6
Wheel 2v	110.55	50	51.56	10.9	11.5	43.5	41.2	79.2	331	H6/r6

Maximum equivalent stresses remain below $R_e/s = 200 \text{ MPa}$ in the reported calculation. Static yielding is therefore not the governing issue. The assembly temperature is.

10.2 Thermal assembly feasibility

The heating estimate adds a $150 \mu\text{m}$ assembly clearance and $5 \mu\text{m}$ surface allowance before applying $\Delta T = a/(\alpha d)$ with $\alpha = 12 \times 10^{-6} \text{ K}^{-1}$. This produces temperature rises from $331 \text{ }^{\circ}\text{C}$ to $591 \text{ }^{\circ}\text{C}$. Such temperatures can oxidise surfaces, alter heat treatment, damage nearby seals and make dimensional control difficult. They are not acceptable as an unqualified workshop instruction.

The design should reduce the required assembly clearance, evaluate hub-only induction heating, consider cooling the shaft, and compare keyed, splined or tapered-lock alternatives. A verified procedure must specify initial temperatures, heating uniformity, maximum material temperature, transfer time, insertion force, final runout and a disassembly strategy. The fit calculation also needs surface roughness correction and a friction coefficient justified by the actual surface and lubricant condition.

11 Lubrication and Thermal Management

11.1 Pitch-line velocity audit

Oil viscosity was initially selected from a load–speed chart. Before using the chart, the pitch-line speed must satisfy the same kinematic value on both members of a mesh:

$$v_p = \frac{\omega d}{2}. \quad (34)$$

For stage u , $v_p = 0.23975 \times 11.4/2 = 1.37$ m/s. For stage v , $v_p = 0.160 \times 41.61/2 = 3.33$ m/s. These values differ materially from the initial calculation values of 10.6 m/s and 1.7 m/s. Figure 17 makes the discrepancy explicit.

The chart abscissa is reconstructed as

$$S_v = \frac{1}{v_p} \frac{F_t}{bd} \frac{u+1}{u} Z_H^2 Z_\varepsilon^2 K_A, \quad (35)$$

with $Z_H^2 Z_\varepsilon^2 = 3$ and $K_A = 1.35$. Using the corrected velocities yields $S_{v,u} \approx 2.30$ MPa s/m and $S_{v,v} \approx 0.99$ MPa s/m. The initial values, 0.30 and 1.94 MPa s/m, were driven by inconsistent pitch speeds; the corresponding viscosity readings of 110 and 260 mm²/s are therefore not design outputs.

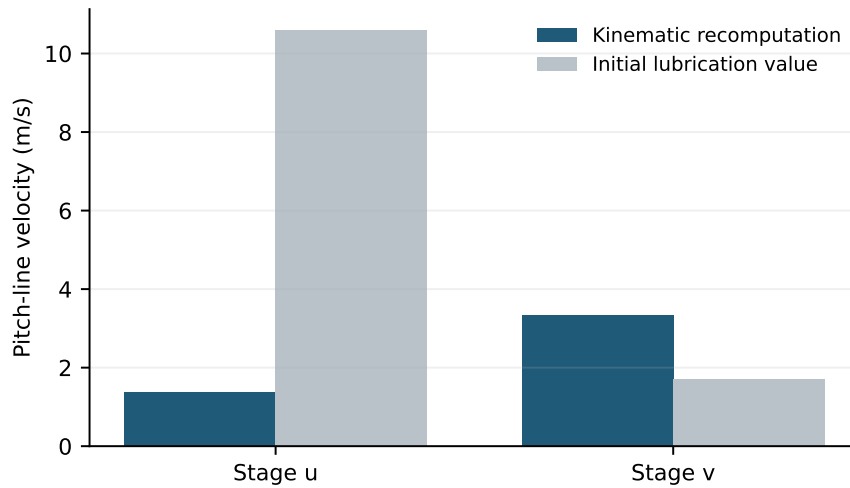


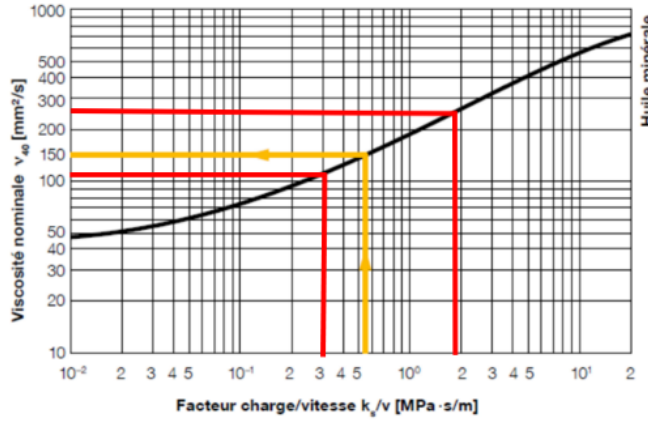
Figure 17 – Pitch-line velocity audit. The viscosity selection must be repeated using the kinematically consistent values.

Because velocity enters the load-speed factor, the chart-derived viscosity of 260 mm²/s cannot be considered validated. This nominal value also lies between common ISO viscosity grades 220 and 320. Final selection should use corrected velocities, operating bulk-oil temperature, viscosity index, start-up temperature, churning loss and micropitting/scuffing requirements [7].

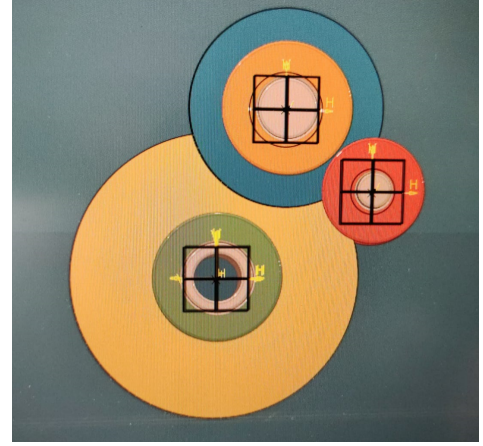
11.2 Splash-lubrication concept

The proposed oil level allows the lower gears to dip into the bath while avoiding full immersion of the meshes. Figure 18 combines the selection chart and the geometric oil-level sketch. This

is a sound concept for moderate pitch speeds, but the housing must include a fill port, drain port, breather, sight glass or level plug, magnetic debris collector and provisions to prevent seal starvation. Oil throw and return paths should be checked on the rotating CAD model.



(a) Original load-speed selection chart.



(b) Proposed splash-oil level.

Figure 18 – Load-speed selection chart and proposed splash-lubrication level.

A thermal balance remains necessary. Mesh, bearing, seal and churning losses generate heat, while the sheet-metal housing rejects heat by natural convection and radiation. The original suggestion of adding external cooling is not yet quantified. A first-principles loss model should be followed by a thermocouple test at nominal power and the most severe ambient condition.

12 Braking, Coupling and Yaw Protection

12.1 Flexible coupling

The generator shaft and gearbox output shaft were specified as 28 mm and 30 mm. The coupling factor is $K = K_1 K_2 K_3 = 1.4 \times 1.0 \times 1.2 = 1.68$, so a nominal output torque of 28.6 N m gives $T_d = KT = 48.0$ N m. A STRAFLEX 635100 flexible coupling was selected. Its 50 N m catalogue limit leaves only about 4% margin over the factored requirement. Misalignment capacity, cyclic torque, start-stop duty and coupling temperature should therefore be verified; a larger size may be justified if the actual generator torque ripple is significant.

12.2 Rotor brake

The emergency brake is mounted on the low-speed rotor shaft. A blade is approximated as a rectangular composite panel with internal cut-out. With outer mass m_e , inner void-equivalent mass $m_i = 0.95m_e$, material density 2500 kg/m^3 , outer dimensions a, b , edge offsets ℓ, e and the parallel centroidal axes used in the CAD model,

$$I_p = \frac{m_e(a^2 + b^2)}{12} - \frac{m_i[(a - 2\ell)^2 + (b - 2e)^2]}{12} = 108 \text{ kg m}^2 \quad (36)$$

per blade. The hollow shaft inertia is 0.09 kg m^2 and the generator inertia is 0.015 kg m^2 . When referred to the rotor side, a high-speed inertia must be multiplied by K_{tot}^2 , not divided by the ratio:

$$J_{\text{eq}} = 3I_p + I_{\text{shaft}} + I_{\text{gen}}K_{\text{tot}}^2 = 327.9 \text{ kg m}^2. \quad (37)$$

The aerodynamic torque used for the two brake cases is obtained from

$$T_a = \frac{\frac{1}{2}\rho AC_p K_r V^3}{\omega}. \quad (38)$$

It gives approximately 504 N m at nominal wind and 1188 N m at the maximum-wind case. The first sizing pass produced required braking torques of 1154.4 N m nominally and 2104.4 N m at maximum wind. Applying the additional fail-safe criterion $T_b \geq 3T_{b,\text{nom}}$ leads to the governing specification

$$T_{b,\text{design}} = \max(2104.4, 3 \times 1154.4) = 3463.2 \text{ N m}. \quad (39)$$

For constant brake and aerodynamic torques, an idealised stopping time is

$$t_s = \frac{J_{\text{eq}}\omega_0}{T_b - T_a}, \quad T_b > T_a. \quad (40)$$

Using $\omega_0 = 15.6 \text{ rad/s}$, $T_a = 1188 \text{ N m}$ and $T_b = 3463 \text{ N m}$ gives $t_s \approx 2.25 \text{ s}$. The initial rotational energy is about 39.9 kJ; this energy, plus work done by the wind during stopping, must be absorbed without fade.

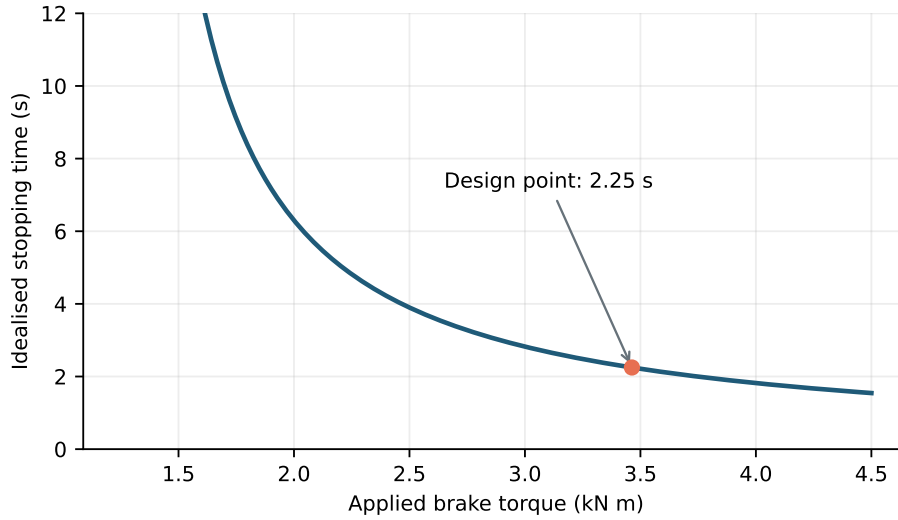


Figure 19 – Idealised emergency stopping time as a function of applied low-speed-shaft brake torque.

The design preference for a normally engaged electromagnetic brake is appropriate for fail-safe stopping after loss of power. A complete safety case must still consider brake release logic, repeated stops, disc temperature, pad wear, overspeed detection, gust torque and the fact that aerodynamic torque does not remain constant during deceleration.

12.3 Yaw bearing and nacelle orientation

The yaw-bearing calculation combines axial gravity load, radial wind force and overturning moment. At 20 m/s,

$$F_r = \frac{1}{2}\rho AC_p V^2 = 1029.6 \text{ N}, \quad (41)$$

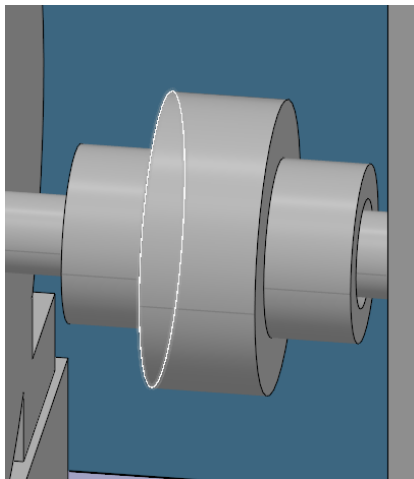
$$F_a = \sum_i m_i g = 5257.34 \text{ N}. \quad (42)$$

Since $F_r/F_a = 0.196 < 0.25$, the catalogue combination factor is $X_r = 0.50$. The unfactored and equivalent bearing loads are therefore

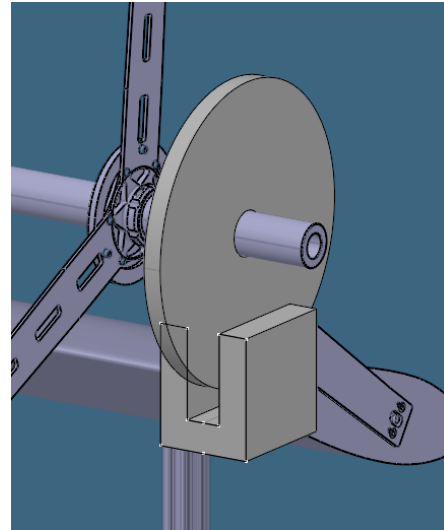
$$F = F_a + X_r F_r = 5772.1 \text{ N}, \quad F_{\text{eq}} = F K_a K_u K_s = 7792.4 \text{ N}, \quad (43)$$

$$M_t = F_r (2.168) = 2232.2 \text{ N m}, \quad M_{t,\text{eq}} = M_t K_a K_u K_s = 3013.4 \text{ N m}, \quad (44)$$

with $K_a = 1.35$, $K_u = 1$ and $K_s = 1$. Candidate slewing rings 03-0217-00 (outer/inner diameters 290/150 mm) and 03-0181-07 (234/125 mm) satisfy the preliminary envelope comparison. Final selection must also check bolt-circle loads, static capacity, overturning stiffness, corrosion protection and yaw-control duty.



(a) Low-speed-shaft brake concept.



(b) Yaw-bearing integration.

Figure 20 – Safety and orientation subsystems integrated in the preliminary CAD model.

13 System-Level Assessment

13.1 Quantified design status

Table 7 separates calculations that close from those that require renewed verification. This prevents a common design-review error: treating the existence of a simulation image or catalogue reference as proof of compliance.

13.2 Sensitivity and uncertainty

The dominant upstream uncertainty is aerodynamic. Because available power scales as V^3 , a 5% wind-speed error changes power by approximately 15.8%. Drivetrain torque at a specified power scales as $1/\omega$, so low-speed operation produces the highest torque. The design should therefore use a load spectrum, not only one nominal state. Turbulence, start-up transients, emergency stopping and grid or load rejection can dominate fatigue even when the mean power is moderate.

The dominant mechanical uncertainty is load-path stiffness. Bearing supports are neither perfectly pinned nor perfectly fixed; housing compliance changes shaft moments, while shaft displacement changes gear load distribution. This coupling explains why a verified gearbox normally uses an iterative shaft-bearing-housing model. The present design workflow recognises that coupling qualitatively, but the numerical models need a common set of coordinates, units and load cases before quantitative correlation is possible.

13.3 Prioritised engineering risks

Table 8 – Prioritised risk register for the next design iteration.

Risk	Likelihood	Consequence	Required closure action
Input-shaft over-stress	Medium–high	High	Audit loads and units; remodel shoulders and fillets; increase diameter or shorten span; perform fatigue check.
Analytical/FEA mismatch	High	High	Rebuild both models in SI units; compare reactions and a benchmark beam case; document mesh convergence.

Risk		Likelihood	Consequence	Required closure action
Gear	misalign- ment	Medium	High	Evaluate relative slope and displacement at each mesh; include housing and bearing stiffness.
Lubricant	mis- selection	High	Medium– high	Recompute pitch velocities and load-speed factors; select a standard ISO VG oil at operating temperature.
Fit assembly	over- heating	High	High	Reduce insertion clearance; design controlled induction/shrink process or use an alternative torque joint.
Brake	thermal fade	Medium	High	Calculate stop energy under gust load; obtain manufacturer energy rating; conduct repeated-stop test.
Bearing	suffix mismatch	Medium	Medium	Verify exact C , C_0 , clearance, seal and speed limits for every reference.
Housing	leak- age/alignment	Medium	Medium– high	Add tolerance datums, gasket compression, oil ports and bore-coaxiality inspection plan.
Unvalidated rotor	curve	Medium	High	Measure or simulate $C_p(\lambda)$, starting torque and extreme-load coefficients.

14 Conclusions and Research Outlook

14.1 Principal findings

The integrated analysis demonstrates that the project has progressed beyond a schematic concept. The power chain, tooth counts, gear diameters, packaging and assembly architecture form a recognisable and largely coherent 4.6 kW wind-turbine drivetrain. The two-stage gearbox closes at a ratio of 13.984, the full belt-plus-gear ratio differs from the original target by less than 1%, and the resulting 1522 rpm output is compatible in magnitude with a four-pole induction machine. The selected gear modules and widths give compact centre distances while carrying tangential mesh loads of 3.37 kN and 1.38 kN. CAD iteration reduces axial gearbox length by 13.8%.

Independent checks also reveal three issues that materially change the design conclusion. First, the bearing-life calculation must convert radians to revolutions; the corrected required dynamic ratings are 8.9–33.1 kN, depending on position. Second, the lubrication table contains pitch-line velocities inconsistent with the gear kinematics, so the viscosity decision must be repeated. Third, and most important, the structural evidence does not validate all shafts. The input shaft exceeds the adopted stress limit and both the input and intermediate shafts exceed the deflection criterion on the reported Abaqus results. The analytical and numerical intermediate-shaft deflections differ by more than four orders of magnitude.

The scientifically defensible conclusion is therefore conditional acceptance of the architecture and rejection of final structural release. The concept is suitable for a new verification iteration, not for manufacture under the present evidence set.

14.2 Recommended verification programme

The next phase should proceed in the following order:

1. Freeze a single requirements and load-case table, including the rotor $C_p(\lambda)$ curve, efficiencies, gust factors, start-stop duty and emergency cases.
2. Rebuild the kinematic, bearing and shaft calculations in one unit-controlled Python workbook and automatically check power, torque and reaction equilibrium.

Table 7 – System-level verification status.

Subsystem	Status	Evidence and next action
Aerodynamic power	Conditional	Power balance closes at $C_p = 0.40$, but $C_p(\lambda)$ and the implicit 0.90 derating factor require aerodynamic validation.
Kinematic chain	Pass	Tooth-count ratio gives 1522 rpm, within 0.9% of the design target.
Gear geometry	Conditional	Geometry and mesh forces close; ISO 6336 material, quality and load-spectrum factors are incomplete.
Bearing life	Conditional	Life was recalculated with correct rad/s-to-rev conversion; exact catalogue ratings and suffixes remain to be verified.
Shaft torsion	Pass at nominal load	Selected diameters meet the $0.5^\circ/\text{m}$ stiffness rule in pure torsion.
Shaft bending / FEA	Fail / unresolved	Input stress and two deflections exceed criteria; analytical and numerical deflections do not correlate.
Housing and support	Conditional	Load path and CAD concept exist; sheet-metal stiffness, fasteners, gasket and bearing-seat alignment require analysis.
Interference fits	Conditional	Static hub stresses pass; heating temperatures are impractical without a controlled assembly process.
Lubrication	Recalculate	The initial pitch-line velocities are inconsistent with gear kinematics.
Emergency brake	Conditional	Ideal stop time is about 2.25 s; thermal capacity and repeated-stop duty are not demonstrated.

3. Correlate a simple analytical beam with Abaqus, then add shoulders, fillets, bearing stiffness and housing compliance. Require displacement and reaction agreement before interpreting peak stress.
4. Complete ISO 6336 contact and tooth-root calculations with a specified steel grade, heat treatment, gear accuracy, reliability and load spectrum.
5. Recalculate lubrication from corrected pitch velocities and perform a gearbox thermal balance.
6. Replace the current shrink-fit heating estimate with a qualified assembly process and tolerance stack.
7. Validate the brake by energy rating and repeated-stop testing, and validate the drivetrain on an instrumented rig before turbine deployment.

This programme would convert the present coherent research prototype into a traceable engineering design whose safety factors, service life and operating temperature are supported by mutually consistent calculations and tests.

A Calculation Traceability

A.1 Power and speed chain

Table 9 – Traceable power and kinematic closure.

Station	ω (rad/s)	n (rpm)	Torque at 4.6 kW (N m)
Rotor, reported	10.10	96.45	455.45
Gearbox input after belt	11.40	108.86	403.51
Intermediate, tooth-count closure	41.10	392.48	111.92
Output, tooth-count closure	159.42	1522.33	28.85
Output, rounded design value	159.80	1525.98	28.79

The rotor torque in this table is the torque corresponding to exactly 4.6 kW. The separate preliminary rotor value 503 N m corresponds to about 5.08 kW; both values are useful, but they represent different stations or loss assumptions and must not be interchanged.

A.2 Bearing life calculation example

For bearing A on the gearbox input shaft,

$$n_e = \frac{11.4 \times 60}{2\pi} = 108.86 \text{ rpm}, \quad (45)$$

$$L_{10,A} = \frac{60 \times 108.86 \times 62400}{10^6} = 407.6 \text{ million revolutions}, \quad (46)$$

$$C_A = 4.0 \times 407.6^{1/3} = 29.7 \text{ kN}. \quad (47)$$

The same procedure is used for B–F. Modified life factors for reliability, contamination, lubrication and material are not included because the required input data are absent.

A.3 Emergency-stop energy

Using the corrected rotor-side inertia from Equation (37),

$$E_k = \frac{1}{2} J_{eq} \omega_0^2 = \frac{1}{2} \times 327.9 \times 15.6^2 = 39.9 \text{ kJ}. \quad (48)$$

This is the initial kinetic energy only. Under constant aerodynamic torque, the wind adds $T_a \theta$ during the stop; a time-domain rotor model is required for a conservative thermal brake specification.

References

- [1] Richard G. Budynas and J. Keith Nisbett. *Shigley's Mechanical Engineering Design*. McGraw-Hill, 11 edition, 2020.
- [2] *Abaqus Analysis User's Guide*. Dassault Systèmes Simulia, 2024.
- [3] DIN. *DIN 7190-1: Interference Fits – Calculation and Design Rules*. DIN, 2017.
- [4] Tedric A. Harris and Michael N. Kotzalas. *Rolling Bearing Analysis*. CRC Press, 5 edition, 2006.
- [5] International Electrotechnical Commission. *IEC 61400-2: Wind Turbines – Part 2: Small Wind Turbines*. International Electrotechnical Commission, 2013.
- [6] International Organization for Standardization. *ISO 281: Rolling Bearings – Dynamic Load Ratings and Rating Life*. International Organization for Standardization, 2007.
- [7] International Organization for Standardization. *ISO/TR 18792: Lubrication of Industrial Gear Drives*. International Organization for Standardization, 2008.
- [8] International Organization for Standardization. *ISO 6336-1: Calculation of Load Capacity of Spur and Helical Gears – Basic Principles, Introduction and General Influence Factors*. International Organization for Standardization, 2019.
- [9] International Organization for Standardization. *ISO 6336-2: Calculation of Load Capacity of Spur and Helical Gears – Calculation of Surface Durability*. International Organization for Standardization, 2019.
- [10] International Organization for Standardization. *ISO 6336-3: Calculation of Load Capacity of Spur and Helical Gears – Calculation of Tooth Bending Strength*. International Organization for Standardization, 2019.
- [11] *Rolling Bearings Catalogue*. SKF Group, 2018.